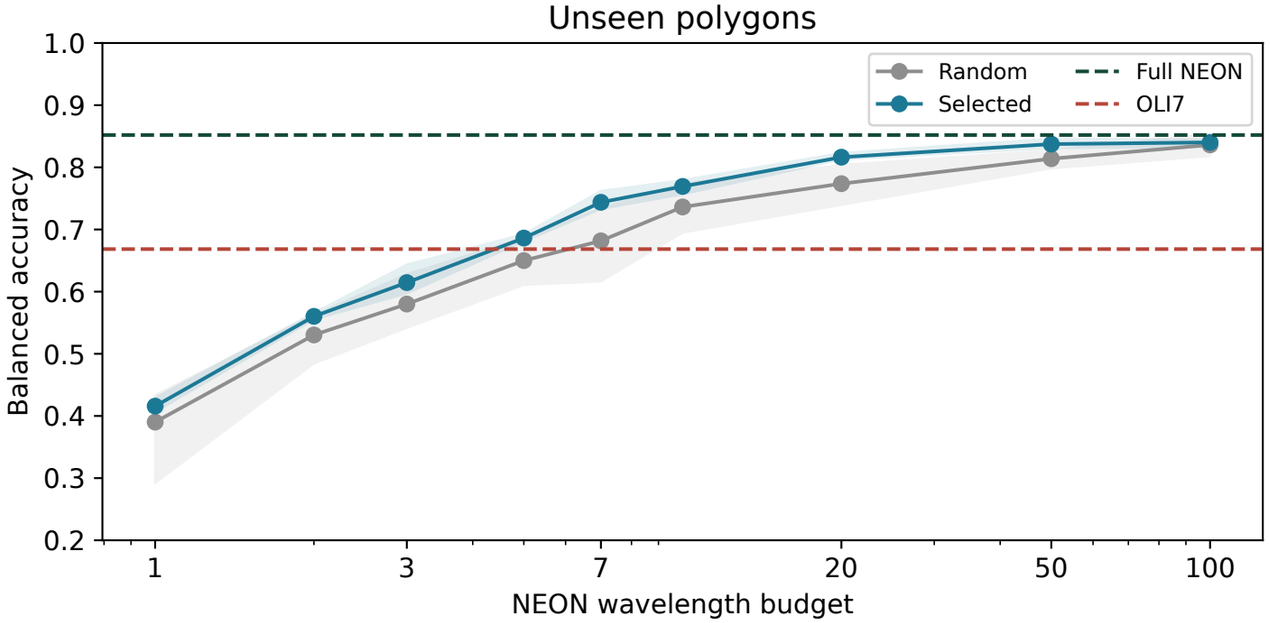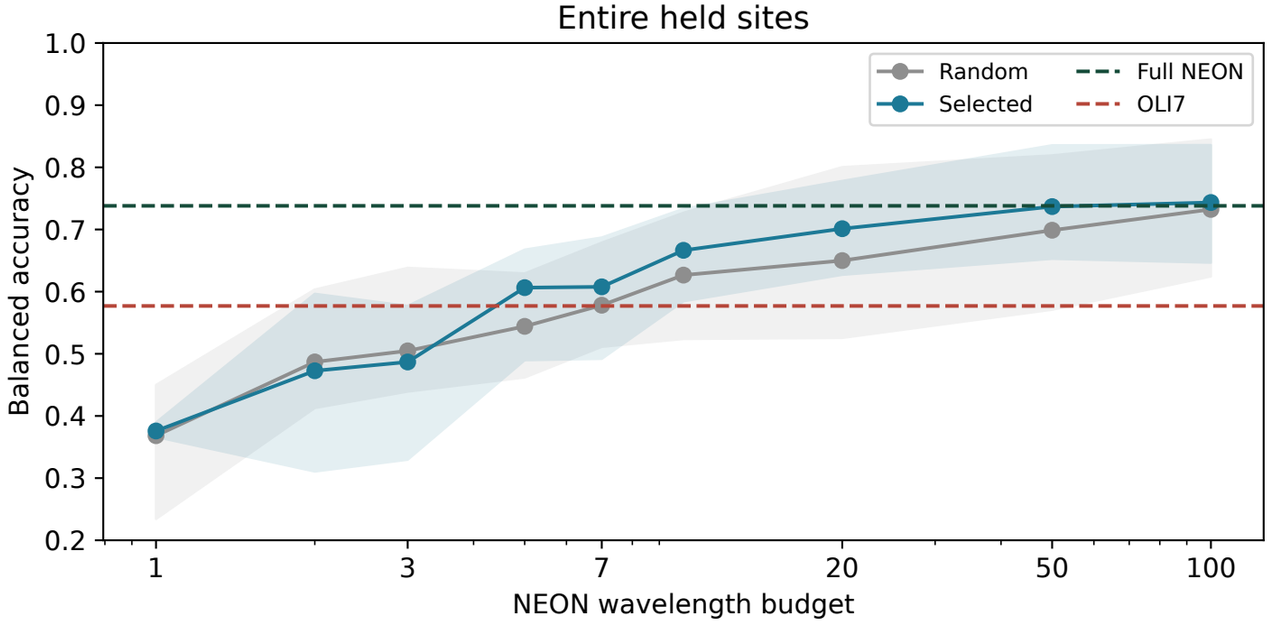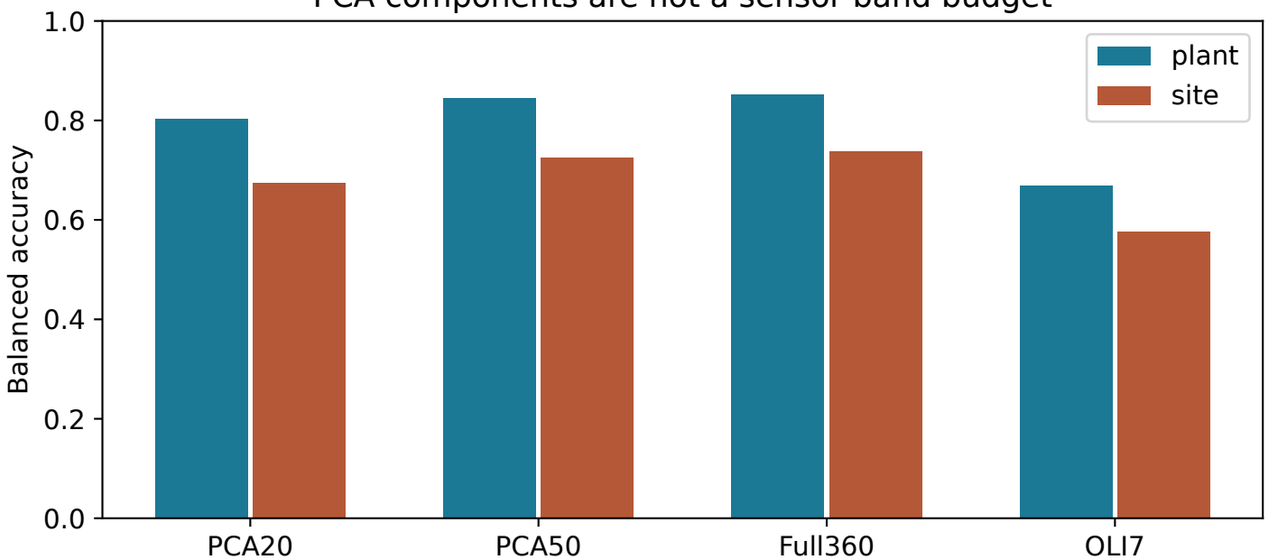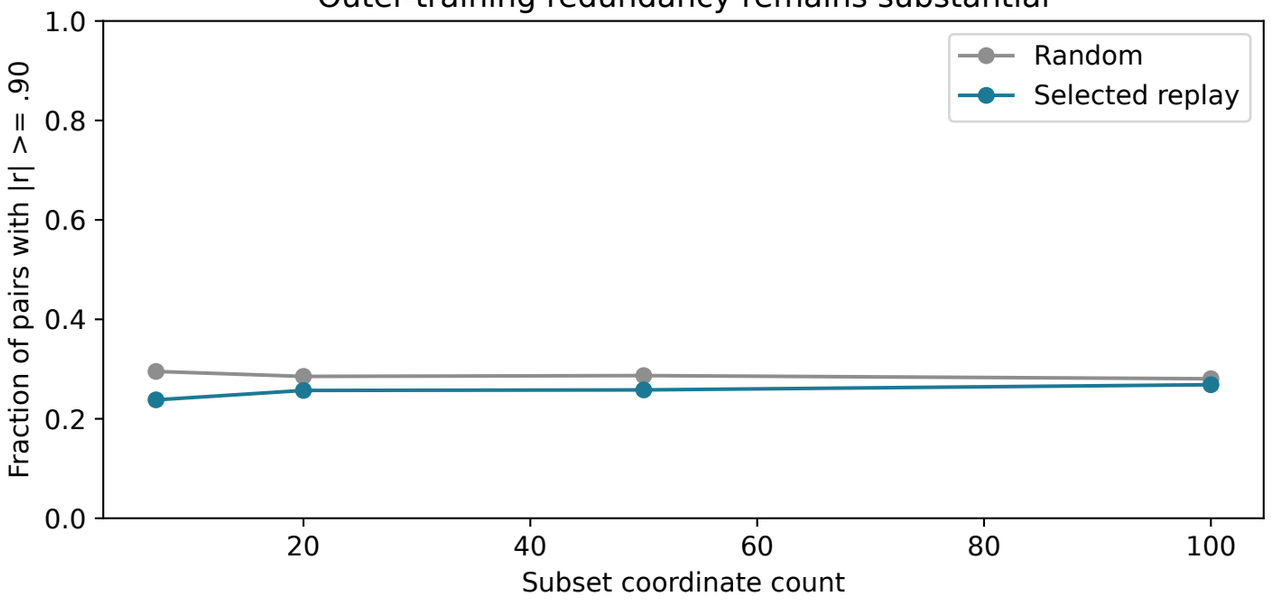Complementary selection adds value, but random bands can catch up